

Characterizing Challenging Behavior in Neurogenetic Syndromes: Exploring Patterns Across Methodologies



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Background

This study aims to examine challenging behavior profiles and families' identified treatment targets through two aims:

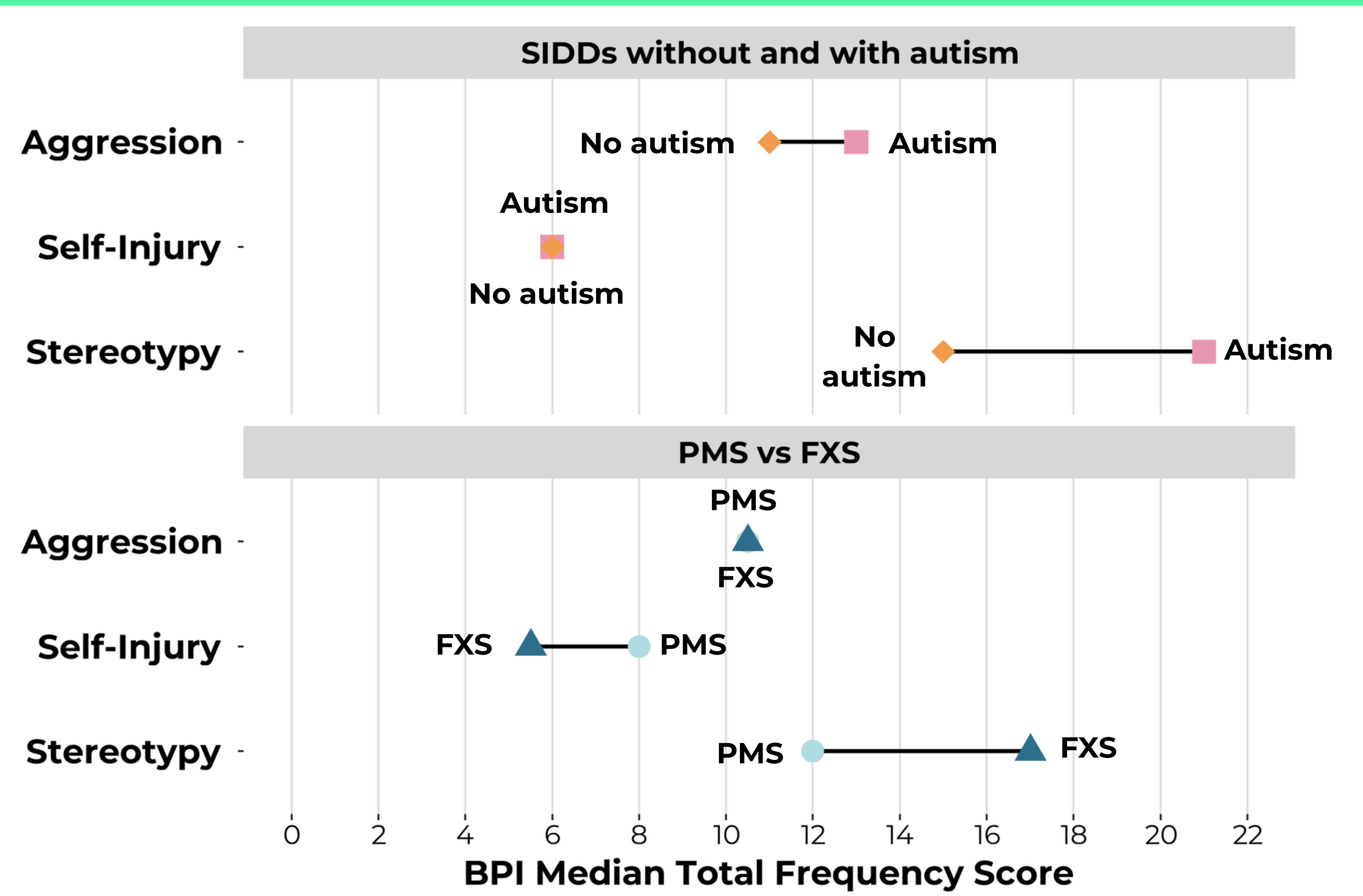
- Describing profiles and family priorities for treatment targets in children with **syndromic intellectual and developmental disabilities (SIDDs) with and without co-occurring autism spectrum disorder**.^{1,2}
- Exploring syndrome-specific behavioral phenotypes which have been reported in the literature for children with **Phelan-McDermid syndrome (PMS)**³ and **Fragile X syndrome (FXS)**.⁴

Note: Both person-first and identity-first language are used interchangeably throughout this poster to support clarity.

Procedure

- 52 caregivers (90.4% female) of eligible participants with SIDDs completed the **Behavior Problems Inventory-01 (BPI-01)**⁵ during clinical screening and the **Parent Target Problems (PTP)**⁶ with an Independent Evaluator (IE) during baseline according to established protocols (NCT06139172).
- PTP** categorization:
 - Two independent reviewers categorized 106 **PTP** labels into 10 topography categories following methodology described in Sheridan et al., 2021.⁷
- Analyses:
 - PTP** scores are presented as the percentage of families ranking the behavior target as priority 1, 2, or 3.
 - Median **BPI-01** subscale total frequency and severity scores across subgroups were calculated.
 - Mann-Whitney *U* tests were applied to evaluate group differences across **BPI-01** frequency and severity.

Figure 1. BPI Median Total Frequency Scores



Median *stereotypy* scores are significantly higher in children with **SIDDs with co-occurring autism**. No other significant group differences were found.

Table 1. Parent Target Problems Across Sample

Group	Parent Target Problem Topography Category, % (n)		
	1	2	3
All (n = 52)	Aggression, 53.8% (28/52)	Compliance Issues, 50% (26/52)	Compliance Issues-Tantrums, 17.3% (9/52)
SIDDs without autism (n = 27)	Aggression, 51.9% (14/27)	Compliance Issues, 48.1% (13/27)	Compliance Issues-Tantrums, 22.2% (6/27)
SIDDs with autism (n = 25)	Aggression, 56% (14/25)	Compliance Issues, 52% (13/25)	Safety Concerns, 24% (6/25)

Table 2. Parent Target Problems in PMS and FXS

Group	Parent Target Problem Topography Category, % (n)		
	1	2	3
PMS (n = 12)	Aggression, 58.3% (7/12)	Compliance Issues, 41.7% (5/12)	Safety Concerns, 33.3% (4/12)
FXS (n = 14)	Aggression, 42.9% (6/14)	Compliance Issues, 57.1% (8/14)	Hyperactivity/Impulsivity, 28.6% (4/14)

The top two caregiver-reported targets were *Aggression* and *Compliance Issues* across all analyses on the **PTP**.

Results

Child Demographics (n = 52)

- Age:** 2-12 years, $M = 7.12$, $SD = 3.25$
- Sex assigned at birth:** Male: $n = 38$, 73.1%; Female: $n = 14$, 26.9%
- Genetic Syndrome:** FXS: $n = 14$, 26.9%; PMS: $n = 12$, 23.1%; Angelman Syndrome (AS): $n = 6$, 11.5%; Other: $n = 20$, 38.5%
- SIDDs with co-occurring autism:** $n = 25$, 48.1%
- Intellectual Disability Level:** Borderline: $n = 3$, 5.8%; Mild: $n = 25$, 48.1%; Moderate: $n = 17$, 32.7%; Severe ID = $n = 7$, 13.5%

BPI-01 Subgroup Mann-Whitney *U* Analyses: Frequency and Severity Median Total Scores

- SIDDs with autism compared to SIDDs without autism**
 - Stereotypy* frequency ($U = 227.5$, $p = .045$) and severity ($U = 193.0$, $p = .051$) scores were marginally higher when autism was co-occurring.
 - No differences in frequency or severity were found in *Self-Injury* (Frequency: $U = 322.0$, $p = .782$; Severity: $U = 191.5$, $p = .169$) or *Aggression* (Frequency: $U = 297.0$, $p = .463$; Severity: $U = 227.5$, $p = .305$).
- PMS and FXS**
 - No group differences were found in frequency or severity across all **BPI-01** domains: *Aggression* (Frequency: $U = 84.0$, $p = 1.0$; Severity: $U = 66.0$, $p = .741$), *Self-Injury* (Frequency: $U = 78.0$, $p = .776$; Severity: $U = 28.0$, $p = .068$), or *Stereotypy* (Frequency: $U = 105.0$, $p = .291$; Severity: $U = 108.0$, $p = .108$).

Discussion

- Caregiver priorities for intervention may not reflect challenging behaviors occurring at the highest severity or frequency.
 - Aggression* and *Compliance* are most often selected as caregiver top concerns for intervention on the **PTP**
 - However, the highest median frequency and severity scores were found on the *Stereotypy* domain of the **BPI-01**
- Comparisons on the **BPI-01** suggest similarities in challenging profiles at the group level in **PMS** and **FXS**.
- Co-occurring autism** is associated with elevated frequency and severity of *Stereotypy* compared to **SIDDs without autism**. However, as noted above, *Stereotypy* is less likely to be selected for intervention targets.



WINGS Website

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References

ClinicalTrials.gov NCT06139172